

MF-21: the regular expansion threshold for every higher-order zero

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This manuscript proves the complete three-part MF-21 target for every integer $m \geq 3$. The conjecture and its prior special cases are due to Barrera, Böttcher, Grudsky, Maximenko, and the cited later authors. The bulk argument uses the simple-loop method developed by Böttcher, Grudsky, Maximenko, and collaborators [BGM10, BG25]. The proof combines that method with an exact finite-matrix inverse-trace calculation and a trace obstruction.

Revision — 19 September 2026 (UTC). Following comments from Sergei Grudsky, this revision supplies the coefficient-uniqueness argument, makes the distinction between the upper estimate (25) and the trace obstruction explicit, proves a sharp lower bound for the maximum error, expands the method attribution, and supplies a direct finite-matrix derivation of the inverse-trace limit. The [revision audit](#) records the checks and the scope of the formal-verification work.

Formal verification — 20 September 2026 (UTC). The [complete Lean formalization](#) verifies Theorem 1, coefficient uniqueness and Corollary 6, including the exact Fourier matrix and one common smooth coefficient family. All five public contracts passed the isolated Linux Comparator and Lean default kernel at the recorded proof revision, with only standard foundational axioms. Separate AI-agent source reviews are retained with the verification evidence.

Review and assistance. The [independent mathematical review](#) and [submission record](#) identify the reviewed source and verification scope. Substantial AI assistance is disclosed. An independent AI-agent audit is informal review; it does not establish external human peer review or formal verification.

1. Exact conclusion and the coefficient functions

Fix an integer $m \geq 3$, and put

$$g(\theta) = (2 - 2 \cos \theta)^m, \quad A_n = T_n(g), \quad h = (n + 2)^{-1}, \quad x_{n,j} = j\pi h.$$

Let the eigenvalues of A_n , in increasing order and counted with multiplicity, be $\lambda_{n,j}$.

Theorem 1. There are real C^∞ functions $d_0, \dots, d_{2m}$ on $[0, \pi]$, with $d_0 = g$, such that all three assertions of MF-21 hold. In particular,

$$\left| \lambda_{n,j} - \sum_{k=0}^p d_k(x_{n,j}) h^k \right| \leq C_p h^{p+1} \quad (0 \leq p \leq 2m - 1, 1 \leq j \leq n) \quad (1)$$

for sufficiently large n ; the same estimate for $p = 2m$ holds for $j \geq \lceil (\log(n + 2))^2 \rceil$; and the latter estimate cannot hold for every j . In fact, no family of continuous coefficient functions on $[0, \pi]$ admits an order- $2m$ expansion with a uniform $O(h^{2m+1})$ remainder over all indices. Constants and lower bounds on n may depend on m, p , but not on n, j .

Here is a constructive definition of the functions. Write

$$\omega_\ell = e^{2\pi i \ell / m}, \quad 1 \leq \ell \leq m - 1.$$

For $0 < \theta \leq \pi$, let $r_\ell(\theta)$ be the root inside the unit circle of

$$2 - r - r^{-1} = \omega_\ell (2 - 2 \cos \theta). \quad (2)$$

Define the real phase by

$$\psi(\theta) = \sum_{\ell=1}^{m-1} \arg(1 - r_\ell(\theta)e^{-i\theta}), \quad \eta(\theta) = \theta + 2\psi(\theta), \quad (3)$$

where each argument is the principal argument in $(-\pi/2, \pi/2)$. Its extension to $\theta = 0$ will be established below. Extend g, η smoothly to a neighborhood of $[0, \pi]$, and, for sufficiently small real h , let $Y(x, h)$ be the solution near x of

$$Y = x + h\eta(Y). \quad (4)$$

The coefficients are

$$d_k(x) = \frac{1}{k!} \left. \frac{\partial^k}{\partial h^k} g(Y(x, h)) \right|_{h=0}. \quad (5)$$

They do not depend on the extension, the matrix size, or the eigenvalue index.

2. Stable roots and a normalized exact boundary determinant

Lemma 2. The functions r_ℓ extend smoothly to $[0, \pi]$, with $r_\ell(0) = 1$. They are nonzero and satisfy, for some $c > 0$,

$$|r_\ell(\theta)| \leq e^{-c\theta} \quad (0 \leq \theta \leq \pi). \quad (6)$$

The phase in (3) is smooth and has endpoint values

$$\psi(0) = \frac{(m-1)\pi}{4}, \quad \psi(\pi) = 0, \quad \eta(0) = \frac{(m-1)\pi}{2}, \quad \eta(\pi) = \pi. \quad (7)$$

Proof. A root on the unit circle in (2) would make its left side real and nonnegative. This is impossible because $\omega_\ell \neq 1$, including the case $\omega_\ell = -1$. The reciprocal quadratic has exactly one root inside. Its roots are simple away from $\theta = 0$: a double root could occur only when the right side is 0 or 4.

Set

$$\kappa_\ell = e^{i(\pi\ell/m - \pi/2)}.$$

The quadratic formula, with its square-root branch chosen near zero, gives

$$r_\ell(\theta) = 1 - \kappa_\ell\theta + O(\theta^2), \quad \Re\kappa_\ell > 0. \quad (8)$$

The square root has a factor θ times an analytic nonvanishing function; this also proves smoothness at zero. The roots never vanish. Compactness, (8), and $|r_\ell| < 1$ away from zero prove (6). All logarithmic derivatives r'_ℓ/r_ℓ are bounded.

For positive θ , every factor in (3) has positive real part. At zero it is θ times a smooth nonzero function, with limit

$$\kappa_\ell + i = 2 \sin\left(\frac{\pi\ell}{2m}\right) e^{i\pi\ell/(2m)}.$$

Thus each argument extends smoothly and its limiting value is $\pi\ell/(2m)$. Summing proves the first value in (7). The multiset of stable roots is closed under conjugation; at $\theta = \pi$, the factors $1 + r_\ell$ occur in conjugate pairs, with any real factor positive. Their principal arguments sum to zero. This proves the remaining statements. $\square$

For $0 < \theta < \pi$, define

$$F_n(\theta) = (n+1)\theta - 2\psi(\theta) = (n+2)\theta - \eta(\theta). \quad (9)$$

Lemma 3. There are real smooth functions E_n on $(0, \pi]$ such that $g(\theta)$ is an eigenvalue of A_n precisely when

$$\sin F_n(\theta) + E_n(\theta) = 0 \quad (0 < \theta < \pi). \quad (10)$$

For constants $C, c > 0$ depending only on m ,

$$|E_n(\theta)| \leq Ce^{-cn\theta}, \quad |E'_n(\theta)| \leq C(n+1)e^{-cn\theta}. \quad (11)$$

Moreover $E_n(\pi) = 0$, and therefore

$$|E_n(\theta)| \leq C(n+1)(\pi - \theta)e^{-cn\pi/2} \quad (\pi/2 \leq \theta \leq \pi). \quad (12)$$

Proof. The Laurent polynomial g is $(2 - z - z^{-1})^m$. At the spectral parameter $g(\theta)$, its $2m$ distinct characteristic roots are

$$\mathcal{Z} = (r_1, \dots, r_{m-1}, z, z^{-1}, q_1, \dots, q_{m-1}), \quad z = e^{i\theta}, \quad q_\ell = r_\ell^{-1}.$$

An eigenvector is a solution of the corresponding order- $2m$ recurrence, with ghost values zero at $1-m, \dots, 0, n+1, \dots, n+m$. Shifting these indices by $m-1$, its boundary determinant is

$$D_n(\theta) = \det \left(\begin{matrix} (w^k)_{0 \leq k \leq m-1, w \in \mathcal{Z}} \\ (w^{n+m+k})_{0 \leq k \leq m-1, w \in \mathcal{Z}} \end{matrix} \right). \quad (13)$$

The ghost-value criterion is equivalent to the finite eigenvalue equation. Indeed, the characteristic roots form a basis for recurrence solutions. A solution vanishing both in the interior and at the ghost values has at least $2m$ consecutive zeros and is identically zero. Hence the kernels of the boundary matrix and the eigenvalue matrix have equal dimensions.

For an ordered root list U , put $V(U) = \prod_{a < b} (U_b - U_a)$. Laplace expansion in the bottom m rows gives

$$D_n = \sum_{|S|=m} \sigma_S V(S) V(S^c) \left(\prod_{w \in S} w \right)^{n+m}, \quad \sigma_S \in \{-1, 1\}, \quad (14)$$

with each sublist in the inherited order. Write $R = (r_1, \dots, r_{m-1})$, $O = (q_1, \dots, q_{m-1})$,

$$Q = \prod_{\ell=1}^{m-1} q_\ell > 0, \quad f = \prod_{\ell=1}^{m-1} (1 - r_\ell z^{-1}) = |f|e^{i\psi}.$$

Exactly two subsets have maximal product modulus Q : $S_+ = \{z\} \cup O$ and $S_- = \{z^{-1}\} \cup O$. Their signs are opposite. For some sign σ , their coefficient products are

$$\sigma V(R) V(O) Q z^{-(m-1)} \bar{f}^2, \quad -\sigma V(R) V(O) Q z^{m-1} f^2. \quad (15)$$

For example, this follows by multiplying $V(z, O) = V(O) \prod (q_\ell - z)$ and $V(R, z^{-1}) = V(R) \prod (z^{-1} - r_\ell)$. Their contribution to (14) is consequently

$$\mathcal{N}_n(\theta) \sin F_n(\theta), \quad \mathcal{N}_n = 2i\sigma V(R) V(O) Q^{n+m+1} |f|^2. \quad (16)$$

This normalizer is nonzero for $0 < \theta \leq \pi$. Every other subset has

$$|b_S(\theta)| := \left| \frac{\prod_{w \in S} w}{Q} \right| \leq \max_{\ell} |r_{\ell}(\theta)| \leq e^{-c\theta}. \quad (17)$$

To see the first inequality, if a subset has both oscillatory roots, it omits at least one exterior root; if it has neither, it contains at least one interior root; and if it has exactly one, it replaces at least one exterior root by an interior root. Each replacement only decreases the modulus further.

Divide every remaining term of (14) by (16). This expresses E_n as a finite sum

$$E_n(\theta) = \sum_{S \neq S_+, S_-} a_S(\theta) b_S(\theta)^{n+m}, \quad a_S = \frac{\sigma_S V(S) V(S^c)}{2i\sigma V(R) V(O) Q |f|^2}. \quad (18)$$

All a_S and their first derivatives are bounded on $[0, \pi]$. At zero, every pairwise root difference has a simple zero: the $2m$ first derivatives of the roots are the distinct numbers $-\kappa_{\ell}, \kappa_{\ell}, i, -i$. Thus both the numerator and denominator of a_S vanish to exactly order $m(m-1)$, with nonzero leading coefficients. Their quotient extends smoothly. At π , the denominator has no zero: the interior roots and the exterior roots are separately distinct, and $f \neq 0$. Some numerators vanish there because $z = z^{-1}$, which causes no singularity.

Every b_S is smooth, nonzero, and has bounded logarithmic derivative. Differentiating (18) and using (17) proves (11), after changing constants. The quotient $D_n/\mathcal{N}_n$ is real for real θ : conjugation permutes the two oscillatory roots, and permutes the interior and exterior lists by the same permutation. Hence $\overline{D_n} = -D_n$, while $V(R)V(O)$, Q , and $|f|^2$ are real, so $\overline{\mathcal{N}_n} = -\mathcal{N}_n$. This proves that E_n is real.

At π , the two oscillatory columns in (13) coincide, so $D_n(\pi) = 0$, whereas $\mathcal{N}_n(\pi) \neq 0$ and $\sin F_n(\pi) = \sin((n+1)\pi) = 0$. Therefore $E_n(\pi) = 0$. Integrating the derivative bound over $[\theta, \pi]$ proves (12). $\square$

3. Root indexing and a uniform exponentially small phase error

All eigenvalues of A_n lie strictly between 0 and 4^m . For nonzero v , its quadratic form is the integral of g times the squared modulus of a nonzero trigonometric polynomial; both g and $4^m - g$ are positive almost everywhere. Hence there are exactly n eigenangles in $(0, \pi)$, counted with multiplicity.

Lemma 4. There is an integer $J \geq 1$, depending only on m , such that for all sufficiently large n and all $J \leq j \leq n$, the j -th eigenangle $\theta_{n,j} = g^{-1}(\lambda_{n,j})$ is simple and satisfies

$$|\theta_{n,j} - Y(x_{n,j}, h)| \leq \frac{Ce^{-cj}}{n+2}, \quad 0 < \theta_{n,j}, Y(x_{n,j}, h) \leq C \frac{j}{n+2}. \quad (19)$$

Proof. Boundedness of η' gives, for large n ,

$$\frac{n+2}{2} \leq F'_n(\theta) \leq 2(n+2), \quad F_n(0) = -\frac{(m-1)\pi}{2}, \quad F_n(\pi) = (n+1)\pi. \quad (20)$$

Let $I_{n,k}$ be the interval on which $|F_n(\theta) - k\pi| \leq \pi/4$, for $J \leq k \leq n$. Choose J large enough. On the whole range $F_n \geq J\pi - \pi/4$, boundedness of η implies $n\theta \geq c_1 J - C_1$. Thus (11) makes $|E_n| < 1/4$ and $|E'_n| < (n+2)/8$ throughout this range, uniformly for all sufficiently large n .

At the two endpoints of $I_{n,k}$, the sine has opposite signs and magnitude $1/\sqrt{2}$. On $I_{n,k}$, its derivative has constant sign and magnitude at least $(n+2)/(2\sqrt{2})$, so the derivative of (10)'s left side has that same sign. There is exactly one root in each $I_{n,k}$, and it is a simple zero of the determinant. The corresponding eigenspace has dimension one: otherwise the boundary matrix in (13) would have nullity at least two, making its determinant derivative zero. Since A_n is symmetric, this eigenvalue also has algebraic multiplicity one.

Between these intervals, and below the final interval $|F_n - (n+1)\pi| \leq \pi/4$, the magnitude of the sine is at least $1/\sqrt{2}$; there are no roots there. The final interval needs a different argument because $\theta = \pi$ is an artificial determinant root. There, $\theta \geq \pi/2$ for large n , and (20) and the elementary lower bound for sine give

$$|\sin F_n(\theta)| \geq c_2(n+2)(\pi - \theta).$$

Equation (12) is strictly smaller for every $\theta < \pi$, once n is large. Thus there is no eigenangle in this final interval.

We have found exactly one simple eigenangle in each interval $I_{n,k}$, $k = J, \dots, n$, and none between or above them. Counting downwards from the largest eigenvalue therefore identifies the root in $I_{n,k}$ as precisely the k -th eigenangle; no assertion about the roots below this range was needed.

At that root, (11) and $|F_n - j\pi| \leq \pi/4$ imply

$$|F_n(\theta_{n,j}) - j\pi| \leq Ce^{-cn\theta_{n,j}} \leq Ce^{-c'j}.$$

Here $n\theta_{n,j} \geq c_3j - C_3$, by boundedness of η . The exact root $F_n(y) = j\pi$ is $y = Y(x_{n,j}, h)$. Applying (20) proves the first bound in (19). The upper bounds follow from $F_n = (n+2)\theta - \eta(\theta)$, boundedness of η , and $j \geq J$. $\square$

We also require elementary estimates covering the few remaining eigenvalues. Let $N = n + 2m$, and let C_N be the circulant matrix with eigenvalues $g(2\pi\ell/N)$, $0 \leq \ell < N$. Its leading $n \times n$ principal submatrix is A_n , because no Fourier coefficient of bandwidth m wraps into that block. Its increasingly ordered eigenvalues are

$$\mu_{N,j} = g\left(\frac{2\pi\lfloor j/2 \rfloor}{N}\right), \quad 1 \leq j \leq N.$$

Cauchy interlacing gives

$$\mu_{N,j} \leq \lambda_{n,j} \leq \mu_{N,j+2m}. \quad (21)$$

Consequently $\lambda_{n,j} \leq C_m(j+2m)^{2m}n^{-2m}$. Also, for $j \geq 2$, using $2\sin(t/2) \geq 2t/\pi$ on $[0, \pi]$,

$$\lambda_{n,j} \geq \left(\frac{4\lfloor j/2 \rfloor}{n+2m}\right)^{2m} \geq \left(\frac{4j}{3(n+2m)}\right)^{2m}. \quad (22)$$

4. The expansions through the claimed threshold

The implicit function theorem applied uniformly on the compact interval in (4) gives smooth Y , so for any fixed p ,

$$g(Y(x, h)) = \sum_{k=0}^p d_k(x)h^k + O(h^{p+1}) \quad (23)$$

uniformly in $x \in [0, \pi]$, with $d_0 = g$. For every fixed integer J , equation (4) and boundedness of η give $Y(\pi jh, h) = O_J(h)$ for $1 \leq j \leq J$. Since g vanishes to order $2m$ at zero, (23) at $p = 2m-1$ therefore gives

$$g(Y(\pi jh, h)) = O_J(h^{2m}), \quad \sum_{k=0}^{2m-1} d_k(\pi jh)h^k = O_J(h^{2m}) \quad (24)$$

uniformly over this fixed finite set of indices. This argument uses the same coefficient family as the bulk expansion.

For $j \geq J$, the mean value theorem, $g'(\theta) = O(\theta^{2m-1})$, and (19) yield

$$|\lambda_{n,j} - g(Y(x_{n,j}, h))| \leq Ch^{2m} j^{2m-1} e^{-cj} \leq Ch^{2m}. \quad (25)$$

For $j < J$, (21) gives $\lambda_{n,j} = O(h^{2m})$, and (24) gives $\sum_{k=0}^{2m-1} d_k(x_{n,j})h^k = O(h^{2m})$, uniformly over this fixed finite set. Combining these observations with (23) at $p = 2m - 1$ proves (1) at that order for every j . Subtracting the omitted terms, whose coefficients are bounded on $[0, \pi]$, proves every lower order in (1), with the same d_k .

If $j \geq \lceil (\log(n+2))^2 \rceil$, the first bound in (25) is $O(h^{2m+1})$: exponential decay beats the displayed polynomial uniformly in that range. Equation (23) with $p = 2m$ now proves Part 2 of the target.

Equation (25) is an upper bound; it does not prove that an $O(h^{2m+1})$ remainder is impossible. That obstruction is proved next by comparing two trace limits, after identifying any hypothetical alternative coefficient family.

5. Coefficient uniqueness and the trace obstruction

The following specialization of the coefficient-uniqueness principle in [BBGM, Proposition 4.2] addresses the distinction emphasized in their Remark 8.3 between failure for a constructed family and failure for every continuous family.

Lemma 5 (uniqueness on the bulk grid). Fix an integer $p \geq 0$, put $h_n = (n+2)^{-1}$ and $L_n = \lceil (\log(n+2))^2 \rceil$, and suppose two real continuous families $d_0, \dots, d_p$ and $e_0, \dots, e_p$ on $[0, \pi]$ satisfy

$$\left| \lambda_{n,j} - \sum_{k=0}^p c_k(j\pi h_n) h_n^k \right| \leq C_c h_n^{p+1} \quad (n \geq N_c, L_n \leq j \leq n),$$

for $c = d$ and $c = e$, with constants independent of n, j . Then $d_k = e_k$ on $[0, \pi]$ for every $0 \leq k \leq p$.

Proof. Induct on k , proving equality on the entire closed interval at each step. Fix $x \in (0, \pi)$ and let $j_n = \lfloor x(n+2)/\pi \rfloor$. Since $j_n/(n+2) \rightarrow x/\pi \in (0, 1)$, $L_n/(n+2) \rightarrow 0$, and $n/(n+2) \rightarrow 1$, eventually $L_n \leq j_n \leq n$. The sample points $x_n = j_n \pi h_n$ lie in $[0, \pi]$ and converge to x .

Subtract the two estimates. If $d_\ell = e_\ell$ on $[0, \pi]$ for every $\ell < k$, the corresponding lower-order terms vanish at x_n . Dividing by $h_n^k > 0$ gives

$$\left| (d_k - e_k)(x_n) + \sum_{\ell=k+1}^p (d_\ell - e_\ell)(x_n) h_n^{\ell-k} \right| \leq (C_d + C_e) h_n^{p+1-k}.$$

The higher coefficients are bounded on the compact interval. Their finite sum tends to zero, as does the right side because $k \leq p$. Continuity therefore gives $d_k(x) = e_k(x)$. Since x was arbitrary in $(0, \pi)$, continuity extends this equality to both endpoints. For $k = 0$ the lower-order hypothesis is empty; thus this proves the base case and the induction. $\square$

Only the highest-order estimates are hypotheses of Lemma 5; separate estimates for the lower truncations are unnecessary. In particular, any continuous family with a global order- $2m$ estimate must coincide on $[0, \pi]$ with the family (5), since both satisfy that estimate on the bulk grid by Section 4.

We derive the inverse-trace limit directly from finite matrices. Write $(a)_q = a(a+1) \cdots (a+q-1)$, with $(a)_0 = 1$, and use zero-based indices for the following inverse column. For $n \geq 1$,

$$u_k^{(n)} := (A_n^{-1})_{k,0} = \binom{m+k-1}{k} \frac{(n-k)_m}{(n+m)_m}, \quad 0 \leq k < n. \quad (26)$$

To verify (26), extend its right side to the polynomial

$$P_n(x) = \frac{(x+1)_{m-1}(n-x)_m}{(m-1)!(n+m)_m}.$$

Its degree is $2m-1$, so its $2m$ -th finite difference vanishes. Also P_n vanishes at the ghost indices $-m+1, \dots, -1$ and $n, \dots, n+m-1$, while $P_n(-m) = (-1)^{m-1}$. The centered difference stencil for A_n has coefficients $(-1)^\ell \binom{2m}{m+\ell}$. For row zero, the omitted leftmost value contributes $(-1)^m P_n(-m) = -1$; all other omitted ghost contributions vanish. Thus $A_n(P_n(0), \dots, P_n(n-1))^T = e_0$, proving (26).

Put $\tau_n = \text{tr}(A_n^{-1})$ and $\tau_0 = 0$. The inverse of the principal tail of A_n is the corresponding block of A_n^{-1} minus $u_{\text{tail}}^{(n)}(u_{\text{tail}}^{(n)})^T/u_0^{(n)}$. This follows by multiplying the two matrices, or by the Schur-complement identity. Taking traces gives the exact recurrence

$$\tau_n - \tau_{n-1} = \frac{\sum_{k=0}^{n-1} (u_k^{(n)})^2}{u_0^{(n)}}. \quad (27)$$

Formula (26) gives $u_0^{(n)} \rightarrow 1$ and the uniform profile limit

$$\max_{0 \leq k < n} \left| n^{1-m} u_k^{(n)} - \frac{(k/n)^{m-1} (1-k/n)^m}{(m-1)!} \right| \rightarrow 0.$$

Indeed, after dividing each rising-factorial factor by n , the numerator is a polynomial in $x = k/n$ and $1/n$, and the denominator tends to $(m-1)! \neq 0$. Riemann sums in (27) therefore yield

$$n^{1-2m}(\tau_n - \tau_{n-1}) \rightarrow \beta_m := \frac{1}{((m-1)!)^2} \int_0^1 x^{2m-2} (1-x)^{2m} dx = \frac{(2m-2)!(2m)!}{(4m-1)!((m-1)!)^2}. \quad (28)$$

Summing the increments, using $n^{-2m} \sum_{k=1}^n k^{2m-1} \rightarrow 1/(2m)$, gives

$$\lim_{n \rightarrow \infty} (n+2)^{-2m} \text{tr}(A_n^{-1}) = \frac{\beta_m}{2m} = \frac{((2m-1)!)^2}{(4m-1)!(2m-1)((m-1)!)^2} \in \mathbb{Q}. \quad (29)$$

This is also the trace constant obtained from the classical Green-kernel limit in Böttcher-Widom [BW, §2]. The finite-matrix derivation above supplies the trace input used here.

Suppose that some continuous coefficient family gave an order- $2m$ estimate uniform for every j . Lemma 5 identifies that family with (5). For each fixed j , (23) then gives

$$\lambda_{n,j} = g(Y(\pi j h, h)) + O(h^{2m+1}).$$

From (4) and (7), $Y(\pi j h, h)/h \rightarrow \pi j + (m-1)\pi/2$. Hence, writing $a = (m-1)/2$,

$$h^{-2m} \lambda_{n,j} \rightarrow \pi^{2m} (j+a)^{2m}. \quad (30)$$

The inverse terms $h^{2m}/\lambda_{n,j}$, extended as zero for $j > n$, are bounded by $C_m j^{-2m}$ for $j \geq 2$, by (22). The $j=1$ term is eventually bounded by (30). Dominated convergence for the counting measure therefore gives a second value for (29):

$$\lim_{n \rightarrow \infty} h^{2m} \text{tr}(A_n^{-1}) = \pi^{-2m} \sum_{j=1}^{\infty} (j+a)^{-2m}. \quad (31)$$

This number is not rational. If $m = 2r + 1 \geq 3$, the sum before multiplication by π^{-2m} is

$$\zeta(2m) - \sum_{\ell=1}^r \ell^{-2m}. \quad (32)$$

If $m = 2r \geq 4$, it is

$$(2^{2m} - 1)\zeta(2m) - \sum_{\ell=0}^{r-1} (\ell + 1/2)^{-2m}. \quad (33)$$

Euler's formula makes $\zeta(2m)/\pi^{2m}$ rational. In both (32) and (33), the subtracted finite sum is a nonzero positive rational number. Thus (31) has the form $u - v/\pi^{2m}$, where $u, v \in \mathbb{Q}$ and $v > 0$. Transcendence of π implies this is irrational. This contradicts (29), proving Part 3 and completing Theorem 1. $\square$

The contradiction does not rely on an interchange of a merely pointwise spectral limit without domination; the uniform tail majorant was supplied explicitly by (22). It also does not infer a general- m statement from the $m = 2$ or $m = 3$ cases.

The same trace comparison yields a lower bound without assuming an improved remainder.

Corollary 6 (sharp maximum-error order). Write

$$P_{n,j} = \sum_{k=0}^{2m} d_k(\pi j h) h^k, \quad R_{2m,n,j} = \lambda_{n,j} - P_{n,j}.$$

There are an integer $J_0 \geq 1$ and constants $c, C > 0$, depending only on m , such that for all sufficiently large $n \geq J_0$,

$$ch^{2m} \leq \max_{1 \leq j \leq J_0} |R_{2m,n,j}| \leq \max_{1 \leq j \leq n} |R_{2m,n,j}| \leq Ch^{2m}.$$

Proof. Put $s = 2m$, $q_{n,j} = h^{-s}\lambda_{n,j}$, and $b_j = \pi^s(j + (m-1)/2)^s$. Let C_m denote the rational value in (29), and let $B_m = \sum_{j=1}^{\infty} b_j^{-1}$. Equations (32)–(33) show that B_m is irrational, independently of any hypothesis on eigenvalue remainders. Thus $\delta = |C_m - B_m| > 0$.

For $j \geq 2$, (22) gives the uniform majorant

$$q_{n,j}^{-1} \leq \left(\frac{3(n+2m)}{4(n+2)j} \right)^s \leq \left(\frac{3m}{4} \right)^s j^{-s} \quad (2 \leq j \leq n).$$

Choose $J_0 \geq 1$ so that both $\sum_{j=J_0+1}^n q_{n,j}^{-1}$, uniformly in $n \geq J_0$, and $\sum_{j=J_0+1}^{\infty} b_j^{-1}$ are less than $\delta/8$. By (29), for all sufficiently large n the full actual sum is within $\delta/8$ of C_m . Choose $\varepsilon > 0$ with $\varepsilon \leq b_1/2$ and $2J_0\varepsilon/b_1^2 \leq \delta/8$.

If $|q_{n,j} - b_j| \leq \varepsilon$ for every $j \leq J_0$, then $q_{n,j} \geq b_j/2$ and

$$|q_{n,j}^{-1} - b_j^{-1}| = \frac{|q_{n,j} - b_j|}{q_{n,j}b_j} \leq \frac{2\varepsilon}{b_1^2}.$$

The difference of the two finite head sums would be at most $\delta/8$. Adding this bound, the two tails, and the trace-limit error would give $\delta \leq \delta/2$, a contradiction. Hence $\max_{j \leq J_0} |q_{n,j} - b_j| > \varepsilon$ for every sufficiently large n .

Equations (4), (7), and (23) give $P_{n,j}/h^s \rightarrow b_j$ for each fixed j , without an eigenvalue-error assumption. Over the finite set $j \leq J_0$ this convergence is uniform, so the last bound implies $\max_{j \leq J_0} |R_{2m,n,j}| \geq (\varepsilon/2)h^s$ eventually. The upper bound follows from (1) at $p = 2m - 1$ and boundedness of d_{2m} . $\square$

Corollary 6 does not require one specified index to attain the lower bound for every large n .

6. Sources, attribution, and verification scope

The target is Conjecture 8.4 of Barrera, Böttcher, Grudsky, and Maximenko [BBGM], p. 26. Their Theorem 1.2 establishes the analogous $m = 2$ threshold. The later seven-diagonal work [BGSV] treats $m = 3$; the 2026 survey [B26], section “Beyond the simple-loop class,” distinguishes these results from broader local expansions. These contributions retain their original attribution.

The simple-loop method originates in the work of Böttcher, Grudsky, and Maximenko [BGM10] and was developed further by these authors and their collaborators. Bogoya-Grudsky [BG25, Theorems 2.1–2.2] give local phase equations and all-order expansions near simple nondegenerate points for general banded Toeplitz matrices. Sections 2–4 specialize the simple-loop approach and provide the estimates and indexing needed here. Deducing the shrinking logarithmic-squared cutoff directly from the local formulation in [BG25] would require control of its constants as the endpoint is approached; no such deduction or novelty claim is asserted here.

The trace constant (29) agrees with the classical inverse-kernel calculation of Böttcher-Widom [BW] and the predecessors credited there; equations (26)–(28) give a direct derivation from finite matrices. Lemma 5 spells out the coefficient-uniqueness principle of [BBGM, Proposition 4.2]. Cauchy interlacing, the finite-dimensional implicit function theorem, Euler’s even-zeta formula, and transcendence of π are standard mathematical inputs. No numerical calculation is a premise of the proof.

The linked submission and review records document the original snapshot and the bounded searches performed for it. The 19 September revision adds [BG25] and expands the proof in response to Grudsky’s comments. No novelty or priority claim is made. The original mathematical reviews were informal AI-agent audits. The revised complete formalization and its independent AI-agent statement reviews are documented above; external human peer review is not asserted.

[BBGM] M. Barrera, A. Böttcher, S. M. Grudsky, and E. A. Maximenko, *Eigenvalues of even very nice Toeplitz matrices can be unexpectedly erratic*, *Operator Theory: Advances and Applications* 268 (2018), 51–77. [arXiv:1710.05243](#), Conjecture 8.4, Theorem 1.2, Proposition 4.2, and Remark 8.3; DOI.

[BGM10] A. Böttcher, S. M. Grudsky, and E. A. Maximenko, *Inside the eigenvalues of certain Hermitian Toeplitz band matrices*, *Journal of Computational and Applied Mathematics* 233 (2010), 2245–2264. DOI.

[BG25] M. Bogoya and S. M. Grudsky, *Eigenvalues of non-Hermitian banded Toeplitz matrices approaching simple points of the limiting set*, *Computational Mathematics and Mathematical Physics* 65 (2025), 1453–1471. Theorems 2.1–2.2; DOI; [author-hosted paper](#).

[BW] A. Böttcher and H. Widom, *From Toeplitz eigenvalues through Green’s kernels to higher-order Wirtinger–Sobolev inequalities*, *Operator Theory: Advances and Applications* 171 (2006), 73–87. [arXiv:math/0412269](#), formula (5), §2 paragraph preceding (13), and (13); DOI.

[BGSV] M. Barrera, S. Grudsky, V. Stukopin, and I. Voronin, *Asymptotics of the eigenvalues of seven-diagonal Toeplitz matrices of a special form*, *Advances in Operator Theory* 9 (2024), 79. [arXiv:2111.07196](#), Theorems 2.3–2.6; DOI.

[B26] A. Böttcher, *Ten years with Sergei Grudsky in the eigenvalue bulk of Toeplitz matrices*, *Journal of Mathematical Sciences* 298 (2026), 363–376. DOI, section “Beyond the simple-loop class.”